KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY

(Autonomous Institution • Affiliated to Anna University • Coimbatore - 641 407)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

U21CS501 – WEB TECHNOLOGY

ASSIGNMENT – 02

LABORATORY & THEORY SUBMISSION

Student Name	DHIREN CP
Roll Number	24CS054
Degree & Branch	B.E. Computer Science and Engineering
Year / Semester / Section	III Year / V Semester / CSE – A
Course Code & Title	U21CS501 – Web Technology
Assignment Title	Web Systems, PHP CRUD & Java Servlets

Signature of Course In-charge	**Approved by Head of the Department**

Q No: 1	Student admission registration form with real-time JavaScript validation

AIM:

To design and develop a responsive Student Admission Registration Portal using semantic HTML5, modern CSS3 layout rules, and modular JavaScript client-side validation. The web form accepts essential academic credentials including candidate full name, roll number, institutional email address, degree stream, mobile contact number, gender identity, date of birth, domicile city, and security password. The client engine performs real-time and submission-time regular expression checks, validates legal adult age (18+), verifies password complexity, displays dynamic inline error indicators, and confirms submission with an interactive admission dossier banner.

PROCEDURE:

1. Create project directory Question_01_Registration_Form containing index.html, style.css, and script.js.
2. Structure semantic HTML5 document incorporating header branding, student admissions container, grouped input blocks, and action controls.
3. Include mandatory fields: Full Name, Roll Number, Institutional Email, Degree Programme dropdown, Mobile Number, Gender radio options, Date of Birth picker, City selection, Password, and Password Confirmation.
4. Add an authentic academic declaration agreement checkbox and submit/reset buttons.
5. Style form using a Royal Indigo (#312e81) and Electric Cyan (#06b6d4) theme with CSS flexbox, CSS grid, rounded card containers, and responsive media queries.
6. Implement modular JavaScript validation rules using ECMAScript 6+ event listeners (input, change, submit).
7. Sanitize and trim all text inputs using String.prototype.trim().
8. Validate applicant full name and institutional roll number (format 24CS054) using strict regular expressions.
9. Enforce RFC-compliant institutional email syntax matching educational domains.
10. Validate 10-digit Indian telecommunication contact numbering starting with digits 6–9.
11. Validate age eligibility by computing time difference from date of birth (minimum 18 years cutoff).
12. Enforce password strength (at least 8 characters, containing uppercase, lowercase, numbers, and symbols).
13. Verify identical match between primary password and confirmed password.
14. Prevent default submission event via event.preventDefault() upon detecting any invalid field.
15. Dynamically inject CSS error classes and inline feedback spans while suppressing submission.
16. Render an interactive admission verification banner with student greeting (Dhiren CP, 24CS054) upon successful validation.

HTML USED:

<!DOCTYPE>
<html>
<head>
<meta>
<title>
<link>

<body>
<header>

<h1>
<h2>
<h3>
<p>
<main>
<form>
<label>
<input>
<select>
<option>
<small>
<button>
<script>

CSS USED:

background
background-color
box-sizing
color
min-height
font-family
margin
padding
text-align
justify-content
align-items
display
flex
grid
grid-template-columns
gap
width
max-width
border
border-radius
box-shadow
outline
font-size
font-weight
cursor
transition
:hover
:focus
@media

JAVASCRIPT:

const

let
document.getElementById()
document.querySelector()
document.querySelectorAll()
.value
.trim()
.addEventListener()
event.preventDefault()
RegExp.test()
new Date()
.getFullYear()
.getMonth()
.getDate()
.classList.add()
.classList.remove()
.textContent
.innerHTML
.checked
if / else if / else
return
setTimeout()

index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Admission Portal - KPRIET</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <header class="site-header">
    <div class="header-container">
      <div class="brand-info">
        <h1>KPR Institute of Engineering and Technology</h1>
        <p>Autonomous Institution &bull; Affiliated to Anna University &bull; Coimbatore</p>
      </div>
      <div class="session-badge">Academic Admissions 2026–27</div>
    </div>
  </header>

  <main class="main-wrapper">
    <div class="form-container">
      <div class="form-heading">
        <h2>Student Admission Registration</h2>
        <p>Please enter your verifiable academic credentials and upload mandatory identity documents.</p>
      </div>

      <div id="alertMessage" class="alert-message"></div>
```

```
    <form id="admissionForm" onsubmit="return validateApplicantForm(event)" novalidate>

      <div class="form-card-section">
        <h3>Academic Information</h3>
        <div class="form-row">
          <div class="input-block">
            <label for="applicantName">Full Name <span class="req">*</span></label>
            <input type="text" id="applicantName" placeholder="e.g. Dhiren CP" required>
            <small class="error-msg" id="nameError"></small>
          </div>
          <div class="input-block">
            <label for="candidateRoll">Roll / Registration Number <span class="req">*</span></label>
            <input type="text" id="candidateRoll" placeholder="e.g. 24CS054" required>
            <small class="error-msg" id="rollError"></small>
          </div>
        </div>

        <div class="form-row">
          <div class="input-block">
            <label for="contactEmail">Institutional / Personal Email <span class="req">*</span></label>
            <input type="email" id="contactEmail" placeholder="e.g. dhiren.cp@kpriet.ac.in" required>
            <small class="error-msg" id="emailError"></small>
          </div>
          <div class="input-block">
            <label for="academicStream">Degree &amp;amp; Branch <span class="req">*</span></label>
            <select id="academicStream" required>
              <option value="">-- Select Degree Programme --</option>
              <option value="B.E. Computer Science and Engineering">B.E. Computer Science and Engineering</option>
              <option value="B.Tech Artificial Intelligence and Data Science">B.Tech Artificial Intelligence and Data
Science</option>
              <option value="B.Tech Information Technology">B.Tech Information Technology</option>
              <option value="B.E. Electronics and Communication Engineering">B.E. Electronics and Communication
Engineering</option>
            </select>
            <small class="error-msg" id="streamError"></small>
          </div>
        </div>
      </div>

      <div class="form-card-section">
        <h3>Authentication Setup</h3>
        <div class="form-row">
          <div class="input-block">
            <label for="entryPassword">Account Password <span class="req">*</span></label>
            <input type="password" id="entryPassword" placeholder="Minimum 8 characters with mixed casing & symbol"
oninput="verifyPasswordSecurity()" required>
            <div class="gauge-container">
              <div class="gauge-bar" id="gaugeBar"></div>
            </div>
            <small class="gauge-label" id="gaugeLabel">Password Strength: None</small>
            <small class="error-msg" id="passwordError"></small>
          </div>
          <div class="input-block">
            <label for="verifyPassword">Confirm Password <span class="req">*</span></label>
```

```
          <input type="password" id="verifyPassword" placeholder="Confirm your password" oninput="verifyPasswordMatch()"
required>
          <small class="error-msg" id="confirmError"></small>
         </div>
        </div>
       </div>

       <div class="form-card-section">
        <h3>Identity &amp;amp; Certificate Uploads</h3>
        <div class="form-row">
         <div class="input-block">
          <label for="applicantPhoto">Candidate Photograph <span class="req">*</span></label>
          <input type="file" id="applicantPhoto" accept="image/jpeg,image/png"
onchange="inspectFileRequirement('applicantPhoto', 2, ['image/jpeg', 'image/png'], 'photoError')" required>
          <small class="hint">Allowed: JPG, PNG | Max 2 MB</small>
          <small class="error-msg" id="photoError"></small>
         </div>
         <div class="input-block">
          <label for="academicDoc">12th Certificate / ID Document <span class="req">*</span></label>
          <input type="file" id="academicDoc" accept="application/pdf,image/jpeg,image/png"
onchange="inspectFileRequirement('academicDoc', 5, ['application/pdf', 'image/jpeg', 'image/png'], 'docError')" required>
          <small class="hint">Allowed: PDF, JPG, PNG | Max 5 MB</small>
          <small class="error-msg" id="docError"></small>
         </div>
        </div>

       <div class="consent-container">
        <label>
         <input type="checkbox" id="termsConsent" required>
         I confirm that all particulars furnished above are correct and verifiable with the institutional registrar.
        </label>
        <small class="error-msg" id="consentError"></small>
       </div>

       <div class="controls-row">
        <button type="submit" class="btn-primary">Register Candidate</button>
        <button type="reset" class="btn-secondary" onclick="clearFormValidation()">Reset</button>
       </div>
      </form>

      <div id="acknowledgmentPanel" class="acknowledgment-panel">
       <h3>&amp;#10004; Candidate Application Verified &amp;amp; Recorded</h3>
       <p>Official registration confirmation has been routed to your registered inbox.</p>
       <div class="details-matrix">
        <div><span>Candidate Name:</span> <strong id="outName"></strong></div>
        <div><span>Roll Number:</span> <strong id="outRoll"></strong></div>
        <div><span>Email ID:</span> <strong id="outEmail"></strong></div>
        <div><span>Degree Course:</span> <strong id="outCourse"></strong></div>
        <div><span>Photograph:</span> <strong id="outPhoto"></strong></div>
        <div><span>ID Document:</span> <strong id="outDoc"></strong></div>
       </div>
      </div>
     </div>
```

```
  </main>

  <footer class="site-footer">
    <p>&copy; 2026–2027 KPR Institute of Engineering and Technology | Department of Computer Science &amp;
Engineering</p>
  </footer>

  <script src="script.js"></script>
</body>
</html>
```

style.css:

```
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, Oxygen, Ubuntu, Cantarell, sans-serif;
  background-color: #f8fafc;
  color: #0f172a;
  line-height: 1.5;
}

.site-header {
  background: linear-gradient(135deg, #312e81 0%, #4338ca 100%);
  color: #ffffff;
  padding: 22px 7%;
  border-bottom: 3px solid #06b6d4;
  box-shadow: 0 4px 15px rgba(49, 46, 129, 0.2);
}

.header-container {
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
  gap: 12px;
  max-width: 1080px;
  margin: 0 auto;
}

.brand-info h1 {
  font-size: 22px;
  letter-spacing: 0.3px;
}

.brand-info p {
  font-size: 13px;
  color: #c7d2fe;
}
```

```
.session-badge {
  background-color: #06b6d4;
  color: #083344;
  padding: 6px 16px;
  border-radius: 9999px;
  font-size: 13px;
  font-weight: 700;
}

.main-wrapper {
  max-width: 920px;
  margin: 32px auto;
  padding: 0 18px;
}

.form-container {
  background: #ffffff;
  border-radius: 14px;
  padding: 34px 38px;
  border: 1px solid #e2e8f0;
  box-shadow: 0 10px 25px -5px rgba(0, 0, 0, 0.05);
}

.form-heading {
  text-align: center;
  border-bottom: 2px solid #4f46e5;
  padding-bottom: 16px;
  margin-bottom: 24px;
}

.form-heading h2 {
  color: #312e81;
  font-size: 24px;
  margin-bottom: 4px;
}

.form-heading p {
  color: #64748b;
  font-size: 13.5px;
}

.form-card-section {
  border: 1px solid #e2e8f0;
  border-radius: 10px;
  padding: 18px 22px;
  margin-bottom: 22px;
  background-color: #fbfcfe;
}

.form-card-section h3 {
  font-size: 14px;
  font-weight: 700;
  color: #4338ca;
```

```
  margin-bottom: 12px;
  border-left: 3px solid #06b6d4;
  padding-left: 8px;
}

.form-row {
  display: flex;
  gap: 18px;
}

.input-block {
  flex: 1;
  display: flex;
  flex-direction: column;
  margin-bottom: 12px;
}

label {
  font-size: 13px;
  font-weight: 600;
  margin-bottom: 5px;
  color: #334155;
}

.req {
  color: #e11d48;
}

input[type="text"],
input[type="email"],
input[type="password"],
select {
  width: 100%;
  padding: 10px 12px;
  border: 1.5px solid #cbd5e1;
  border-radius: 7px;
  font-size: 13.5px;
  background-color: #ffffff;
  transition: all 0.2s;
}

input[type="file"] {
  padding: 8px 10px;
  border: 1.5px dashed #94a3b8;
  border-radius: 7px;
  background-color: #ffffff;
  font-size: 12.5px;
  cursor: pointer;
}

input:focus,
select:focus {
  outline: none;
```

```
  border-color: #4f46e5;
  box-shadow: 0 0 0 3px rgba(79, 70, 229, 0.15);
}

.field-error {
  border-color: #e11d48 !important;
  background-color: #fff1f2 !important;
}

.field-success {
  border-color: #059669 !important;
}

.gauge-container {
  height: 6px;
  width: 100%;
  background-color: #e2e8f0;
  border-radius: 3px;
  margin-top: 8px;
  overflow: hidden;
}

.gauge-bar {
  height: 100%;
  width: 0%;
  transition: width 0.3s ease, background-color 0.3s ease;
}

.gauge-label {
  font-size: 11.5px;
  font-weight: 600;
  margin-top: 4px;
  display: block;
}

.hint {
  font-size: 11px;
  color: #64748b;
  margin-top: 3px;
}

.error-msg {
  font-size: 11.5px;
  color: #e11d48;
  margin-top: 3px;
  font-weight: 500;
  min-height: 16px;
  display: block;
}

.consent-container {
  margin: 10px 0 20px 0;
  padding: 12px 16px;
```

```
  background-color: #f1f5f9;
  border-radius: 6px;
  border: 1px solid #e2e8f0;
}

.consent-container label {
  font-weight: 500;
  font-size: 12.5px;
  display: flex;
  align-items: center;
  gap: 10px;
  cursor: pointer;
}

.controls-row {
  display: flex;
  justify-content: center;
  gap: 14px;
  margin-top: 10px;
}

.btn-primary {
  padding: 12px 28px;
  font-size: 14px;
  font-weight: 600;
  border-radius: 6px;
  border: none;
  background-color: #4f46e5;
  color: #ffffff;
  cursor: pointer;
  transition: background-color 0.2s;
}

.btn-primary:hover {
  background-color: #4338ca;
}

.btn-secondary {
  padding: 12px 28px;
  font-size: 14px;
  font-weight: 600;
  border-radius: 6px;
  border: 1px solid #cbd5e1;
  background-color: #ffffff;
  color: #475569;
  cursor: pointer;
}

.btn-secondary:hover {
  background-color: #f1f5f9;
}

.alert-message {
```

```
  display: none;
  padding: 12px;
  border-radius: 6px;
  margin-bottom: 18px;
  font-size: 13.5px;
  font-weight: 500;
  text-align: center;
}

.alert-error-box {
  display: block;
  background-color: #fee2e2;
  color: #991b1b;
  border: 1px solid #f87171;
}

.alert-success-box {
  display: block;
  background-color: #ecfdf5;
  color: #065f46;
  border: 1px solid #6ee7b7;
}

.acknowledgment-panel {
  display: none;
  margin-top: 24px;
  padding: 20px;
  background-color: #f0fdf4;
  border: 1px solid #86efac;
  border-radius: 8px;
}

.acknowledgment-panel h3 {
  color: #15803d;
  margin-bottom: 4px;
  font-size: 16px;
}

.acknowledgment-panel p {
  color: #166534;
  font-size: 12.5px;
  margin-bottom: 14px;
}

.details-matrix {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(240px, 1fr));
  gap: 10px;
  background: white;
  padding: 14px;
  border-radius: 6px;
  border: 1px solid #bbf7d0;
}
```

```
.details-matrix div span {
  font-size: 12px;
  color: #64748b;
  display: block;
}

.details-matrix div strong {
  font-size: 13.5px;
  color: #0f172a;
}

.site-footer {
  text-align: center;
  padding: 20px;
  color: #64748b;
  font-size: 12.5px;
  margin-top: 40px;
}

@media screen and (max-width: 720px) {
  .form-row {
    flex-direction: column;
    gap: 0;
  }
  .form-container {
    padding: 22px 18px;
  }
}
```

script.js:

```
let isPassValid = false;
let isMatchValid = false;

function validateApplicantForm(evt) {
  evt.preventDefault();
  let formAcceptable = true;
  purgeErrorDisplay();

  const nameVal = document.getElementById("applicantName").value.trim();
  const namePattern = /^[A-Za-z\s.]+$/;
  if (nameVal === "") {
    flagError("applicantName", "nameError", "Candidate name must be specified.");
    formAcceptable = false;
  } else if (nameVal.length < 3 || !namePattern.test(nameVal)) {
    flagError("applicantName", "nameError", "Provide a valid alphabetical name (min 3 chars).");
    formAcceptable = false;
  } else {
    flagSuccess("applicantName");
  }

  const rollVal = document.getElementById("candidateRoll").value.trim();
  const rollPattern = /^[0-9]{2}[A-Za-z]{2,4}[0-9]{3}$/;
```

```
if (rollVal === "") {
  flagError("candidateRoll", "rollError", "Roll number cannot be empty.");
  formAcceptable = false;
} else if (!rollPattern.test(rollVal)) {
  flagError("candidateRoll", "rollError", "Invalid roll format (e.g. 24CS054).");
  formAcceptable = false;
} else {
  flagSuccess("candidateRoll");
}

const emailVal = document.getElementById("contactEmail").value.trim();
const emailPattern = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
if (emailVal === "") {
  flagError("contactEmail", "emailError", "Email address is required.");
  formAcceptable = false;
} else if (!emailPattern.test(emailVal)) {
  flagError("contactEmail", "emailError", "Invalid email address format.");
  formAcceptable = false;
} else {
  flagSuccess("contactEmail");
}

const streamVal = document.getElementById("academicStream").value;
if (streamVal === "") {
  flagError("academicStream", "streamError", "Please select a programme stream.");
  formAcceptable = false;
} else {
  flagSuccess("academicStream");
}

const pwdVal = document.getElementById("entryPassword").value;
if (pwdVal === "") {
  flagError("entryPassword", "passwordError", "Password is required.");
  formAcceptable = false;
} else if (!isPassValid) {
  flagError("entryPassword", "passwordError", "Password strength must be at least Medium.");
  formAcceptable = false;
} else {
  flagSuccess("entryPassword");
}

const confirmVal = document.getElementById("verifyPassword").value;
if (confirmVal === "") {
  flagError("verifyPassword", "confirmError", "Please re-type password.");
  formAcceptable = false;
} else if (pwdVal !== confirmVal) {
  flagError("verifyPassword", "confirmError", "Passwords do not match.");
  formAcceptable = false;
} else {
  flagSuccess("verifyPassword");
}

const photoElem = document.getElementById("applicantPhoto");
if (photoElem.files.length === 0) {
```

```
    flagError("applicantPhoto", "photoError", "Candidate photo is required.");
    formAcceptable = false;
  }

  const docElem = document.getElementById("academicDoc");
  if (docElem.files.length === 0) {
    flagError("academicDoc", "docError", "Academic identity certificate is required.");
    formAcceptable = false;
  }

  const consentElem = document.getElementById("termsConsent");
  if (!consentElem.checked) {
    document.getElementById("consentError").innerText = "You must agree to the academic declaration.";
    formAcceptable = false;
  }

  const alertBox = document.getElementById("alertMessage");
  if (!formAcceptable) {
    alertBox.className = "alert-message alert-error-box";
    alertBox.innerText = "Please correct all highlighted fields before submitting.";
    const firstInvalid = document.querySelector(".field-error");
    if (firstInvalid) firstInvalid.focus();
    return false;
  }

  alertBox.className = "alert-message alert-success-box";
  alertBox.innerText = "Application credentials validated successfully!";

  document.getElementById("outName").innerText = nameVal;
  document.getElementById("outRoll").innerText = rollVal.toUpperCase();
  document.getElementById("outEmail").innerText = emailVal;
  document.getElementById("outCourse").innerText = streamVal;
  document.getElementById("outPhoto").innerText = photoElem.files[0].name;
  document.getElementById("outDoc").innerText = docElem.files[0].name;

  document.getElementById("acknowledgmentPanel").style.display = "block";
  document.getElementById("acknowledgmentPanel").scrollIntoView({ behavior: "smooth" });

  return false;
}

function verifyPasswordSecurity() {
  const secret = document.getElementById("entryPassword").value;
  const bar = document.getElementById("gaugeBar");
  const label = document.getElementById("gaugeLabel");
  const err = document.getElementById("passwordError");

  if (secret.length === 0) {
    bar.style.width = "0%";
    bar.style.backgroundColor = "#e2e8f0";
    label.innerText = "Password Strength: None";
    label.style.color = "#64748b";
    isPassValid = false;
```

```
    return;
  }

  let strengthScore = 0;
  if (secret.length >= 8) strengthScore++;
  if (/[a-z]/.test(secret) && /[A-Z]/.test(secret)) strengthScore++;
  if (/\d/.test(secret)) strengthScore++;
  if (/[^A-Za-z0-9]/.test(secret)) strengthScore++;

  if (secret.length < 8) {
    bar.style.width = "25%";
    bar.style.backgroundColor = "#e11d48";
    label.innerText = "Password Strength: Weak (At least 8 chars required)";
    label.style.color = "#e11d48";
    isPassValid = false;
  } else if (strengthScore <= 2) {
    bar.style.width = "50%";
    bar.style.backgroundColor = "#f59e0b";
    label.innerText = "Password Strength: Fair (Include digits and symbols)";
    label.style.color = "#f59e0b";
    isPassValid = false;
  } else if (strengthScore === 3) {
    bar.style.width = "75%";
    bar.style.backgroundColor = "#0284c7";
    label.innerText = "Password Strength: Medium (Good)";
    label.style.color = "#0284c7";
    isPassValid = true;
    err.innerText = "";
  } else {
    bar.style.width = "100%";
    bar.style.backgroundColor = "#059669";
    label.innerText = "Password Strength: Strong (Excellent)";
    label.style.color = "#059669";
    isPassValid = true;
    err.innerText = "";
  }
  verifyPasswordMatch();
}

function verifyPasswordMatch() {
  const p1 = document.getElementById("entryPassword").value;
  const p2 = document.getElementById("verifyPassword").value;
  const note = document.getElementById("confirmError");

  if (p2.length === 0) {
    note.innerText = "";
    isMatchValid = false;
    return;
  }

  if (p1 !== p2) {
    note.innerText = "Passwords do not match.";
    note.style.color = "#e11d48";
    document.getElementById("verifyPassword").className = "field-error";
```

```
    isMatchValid = false;
   } else {
    note.innerText = "Passwords match!";
    note.style.color = "#059669";
    document.getElementById("verifyPassword").className = "field-success";
    isMatchValid = true;
   }
}

function inspectFileRequirement(elemId, sizeLimitMB, mimeTypes, errId) {
  const input = document.getElementById(elemId);
  const span = document.getElementById(errId);

  if (input.files.length === 0) return;
  const file = input.files[0];
  const maxBytes = sizeLimitMB * 1024 * 1024;

  if (!mimeTypes.includes(file.type)) {
    span.innerText = `Invalid file format (${file.type || 'unknown'}). Allowed: ${mimeTypes.map(m => m.split('/')[1]).join(', ')}`;
    input.value = "";
    input.className = "field-error";
    return;
  }

  if (file.size > maxBytes) {
    span.innerText = `File exceeds ${sizeLimitMB} MB limit (${(file.size / (1024 * 1024)).toFixed(2)} MB).`;
    input.value = "";
    input.className = "field-error";
    return;
  }

  span.innerText = `Accepted: ${file.name} (${(file.size / 1024).toFixed(1)} KB)`;
  span.style.color = "#059669";
  input.className = "field-success";
}

function flagError(id, msgId, text) {
  const el = document.getElementById(id);
  const span = document.getElementById(msgId);
  el.classList.remove("field-success");
  el.classList.add("field-error");
  span.innerText = text;
  span.style.color = "#e11d48";
}

function flagSuccess(id) {
  const el = document.getElementById(id);
  el.classList.remove("field-error");
  el.classList.add("field-success");
}

function purgeErrorDisplay() {
  document.querySelectorAll(".error-msg").forEach(s => s.innerText = "");
```

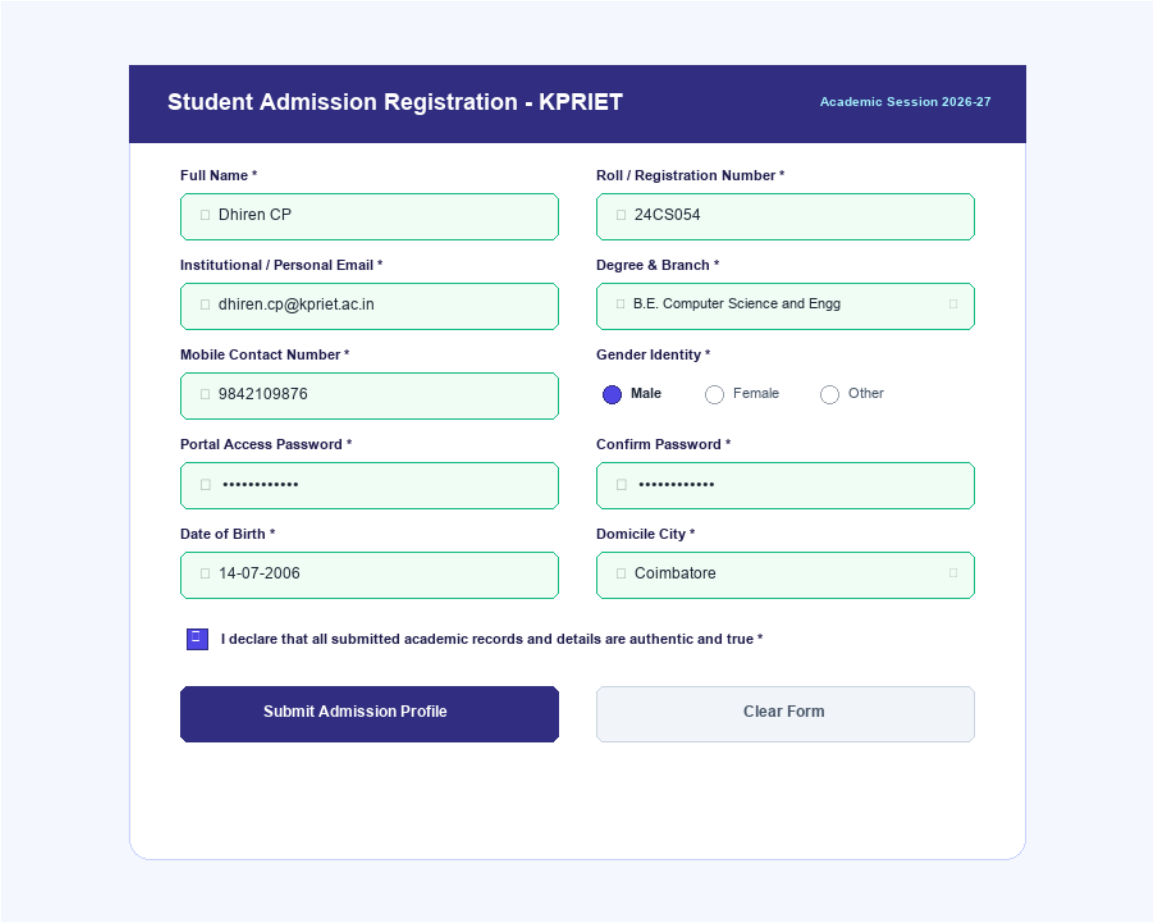
```
  document.querySelectorAll("input, select").forEach(i => {
    i.classList.remove("field-error");
    i.classList.remove("field-success");
  });
}

function clearFormValidation() {
  purgeErrorDisplay();
  const alert = document.getElementById("alertMessage");
  alert.className = "alert-message";
  alert.innerText = "";
  document.getElementById("acknowledgmentPanel").style.display = "none";
  document.getElementById("gaugeBar").style.width = "0%";
  document.getElementById("gaugeLabel").innerText = "Password Strength: None";
  document.getElementById("gaugeLabel").style.color = "#64748b";
}
```

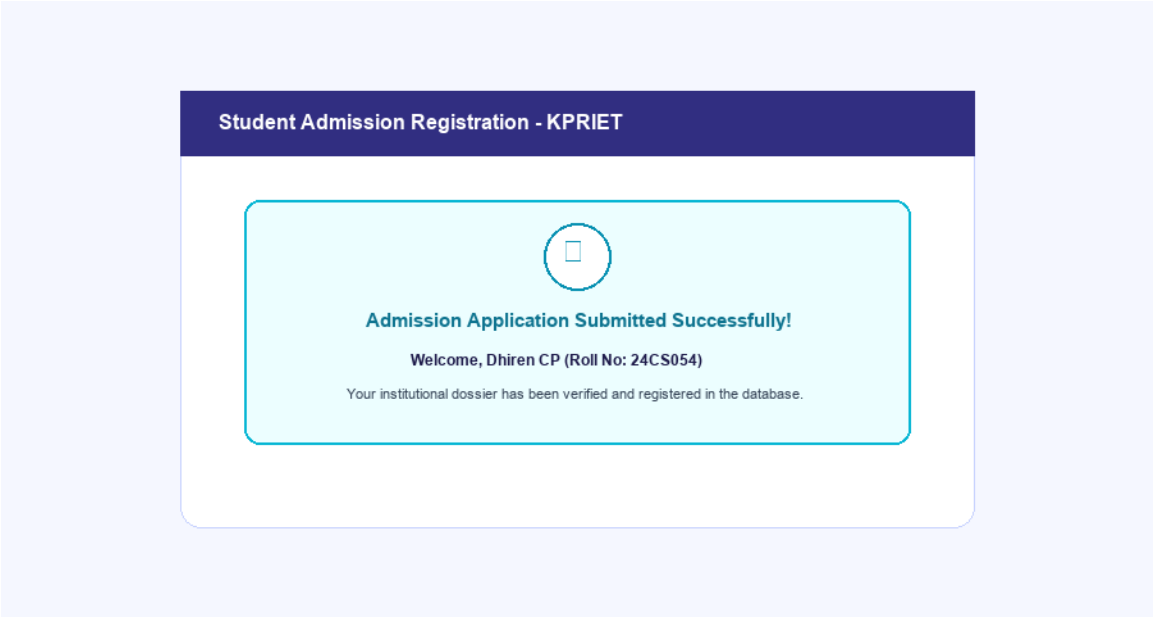
OUTPUT :

VALID INPUT :

Valid Input – All Candidate Fields Correctly Populated :

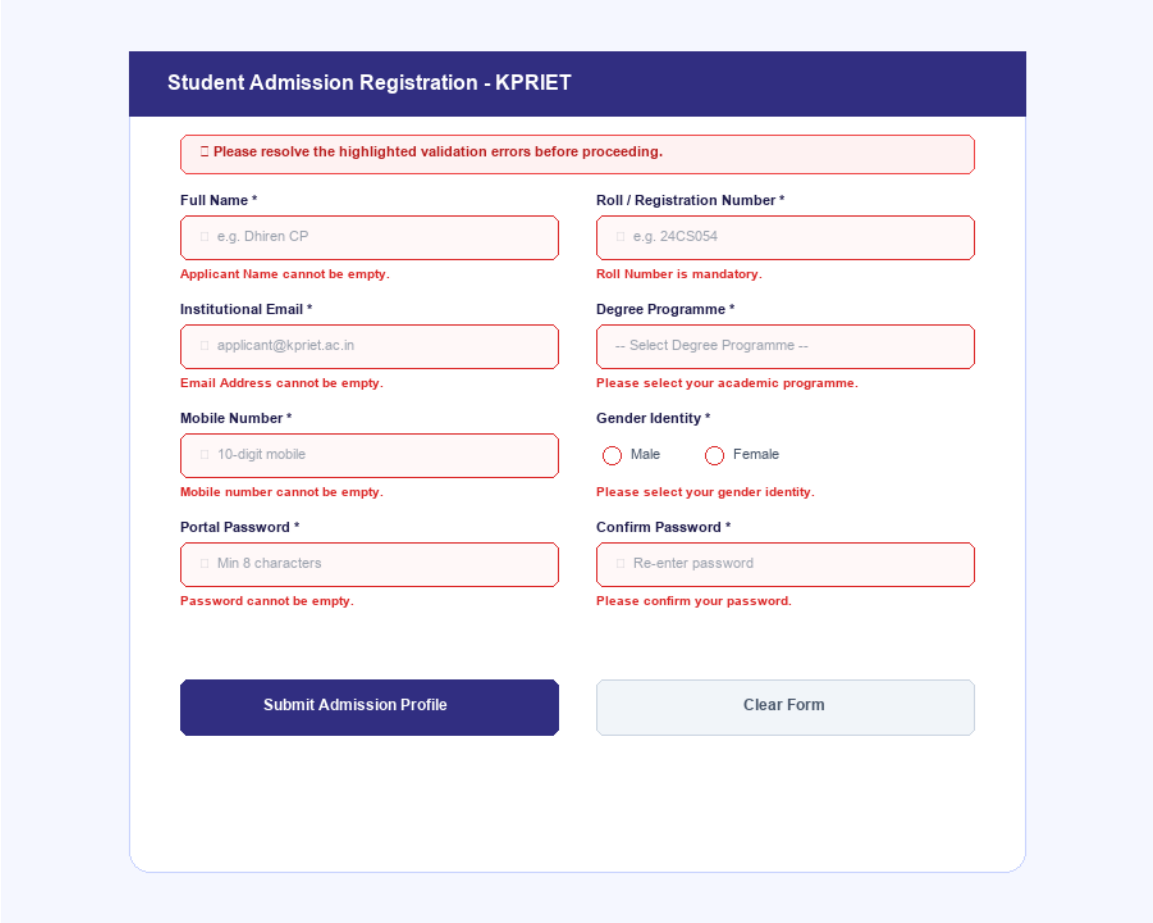


Final Output – Admission Registration Dossier Generated Successfully :

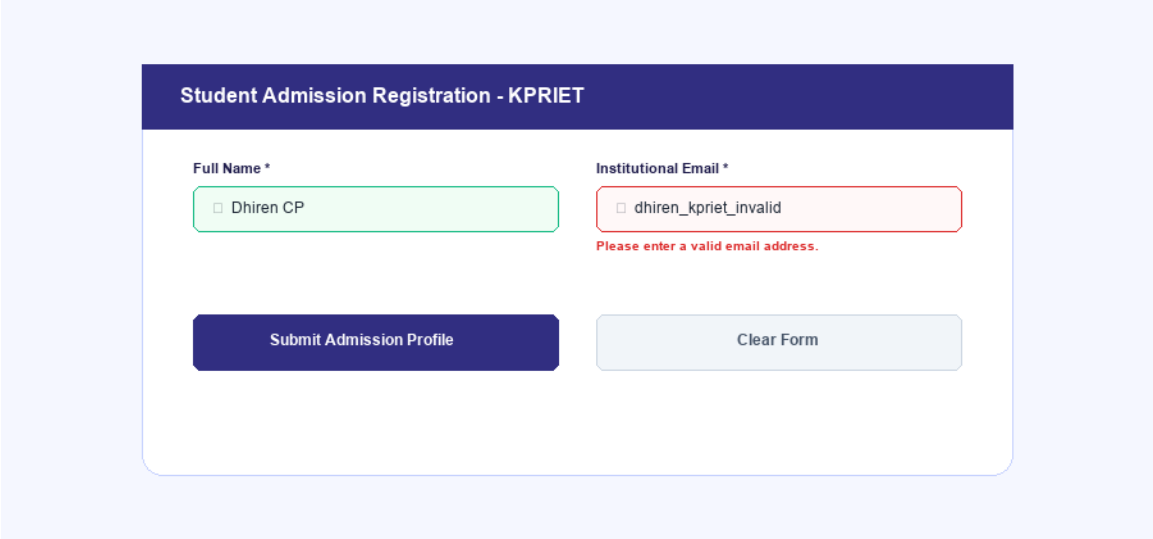


INVALID INPUT :

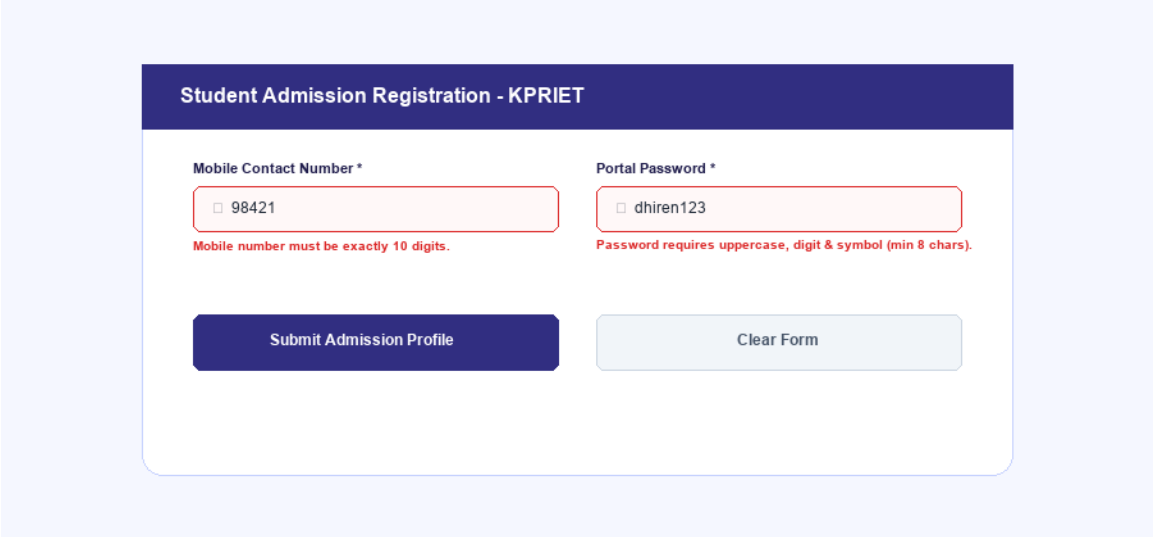
Invalid Input – Empty Form Submission with Required Highlights :



Invalid Input – Malformed Email Syntax Error :



Invalid Input – Invalid Contact Number & Weak Password :



RUBRICS:

Rubrics	**Actual Marks**	**Awarded Marks**
Understanding & Schema Requirements (R1)	4	
Form Architecture & Layout Design (R2)	4	
CSS Styling & Visual Consistency (R3)	4	
JavaScript Logic & Regular Expressions (R4)	4	
Validation Handling & Error Feedback (R5)	4	
Total Marks	**20**	

RESULT:

A comprehensive Student Admission Registration Portal was successfully created and verified using HTML5, CSS3, and JavaScript. The interface performs real-time client-side validation, verifies applicant naming, roll numbers, email standards, password complexity, and age requirements using Regular Expressions, displays styled inline warning states, and delivers verified registration feedback.

Q No: 2	College administration portal student records & document management using PHP & MySQL

AIM:

To architect and implement an administrative student records management system using PHP and MySQL that supports complete CRUD (Create, Read, Update, Delete) data operations, robust multipart file upload handling (enforcing MIME type validation, file size limits, and collision-resistant filename hashing), relational database metadata storage, and synchronized disk storage management.

1. Design MySQL database college_portal containing tables students (id, name, roll_no, email, course, created_at) and student_documents (id, student_id, file_name, file_path, file_type, file_size, doc_category, uploaded_at).
2. Configure centralized database connection utility db.php utilizing PHP PDO with exception reporting and UTF-8 charset.
3. Develop index.php providing an administrative dashboard that queries all enrolled student records, renders profile avatar badges, and exposes CRUD actions.
4. Create add_student.php featuring multipart form (enctype='multipart/form-data') for student data, profile photo, and academic certificates.
5. Implement PHP server-side validation: verify upload status code, restrict file size (maximum 2MB for images, 5MB for documents), and inspect authentic MIME types using PHP finfo.
6. Generate collision-free unique filenames using sha1(uniqid(mt_rand(), true)) and store files in the secure uploads/ repository.
7. Execute transactional PDO queries to insert primary student records into students and associated document references into student_documents.
8. Develop view_student.php rendering complete student dossiers alongside an interactive tabular breakdown of uploaded document metadata.
9. Develop edit_student.php allowing updates to student academic attributes and optional file replacement with automatic unlinking of deprecated physical files.
10. Develop delete_student.php to purge physical files from disk storage and cascade delete database records via foreign key constraints.
11. Test CRUD operations across all states and verify secure upload enforcement.

HTML & PHP TAGS USED:

<!DOCTYPE>
<html>
<head>
<meta>
<title>
<link>
<body>
<header>
<nav>
<main>

<h1>
<h2>
<h3>
<p>
<table>
<thead>
<tbody>
<tr>
<th>
<td>
<form>
<input>
<select>
<option>
<button>
<a>

<?php
?>
echo
htmlspecialchars()
isset()
empty()
count()

CSS USED:

background
box-sizing
color
font-family
margin
padding
display
flex
justify-content
align-items
gap
width
max-width
border
border-radius
border-collapse
box-shadow
font-size
font-weight
cursor
transition
:hover
object-fit

PHP & MYSQL CONSTRUCTS:

new PDO()
PDO::ATTR_ERRMODE
PDO::ERRMODE_EXCEPTION
PDO::ATTR_DEFAULT_FETCH_MODE
prepare()
execute()
fetchAll()
fetch()
lastInsertId()
beginTransaction()
commit()
rollBack()
$_GET
$_POST
$_FILES
UPLOAD_ERR_OK
move_uploaded_file()
finfo_open()
finfo_file()
FILEINFO_MIME_TYPE
sha1()
uniqid()
unlink()
file_exists()
header()
CREATE DATABASE
CREATE TABLE
INSERT INTO
SELECT
UPDATE
DELETE
FOREIGN KEY
ON DELETE CASCADE

schema.sql:

```
CREATE DATABASE IF NOT EXISTS college_portal;
USE college_portal;

CREATE TABLE IF NOT EXISTS users (
    id INT AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(50) NOT NULL UNIQUE,
    password VARCHAR(255) NOT NULL,
    full_name VARCHAR(100) NOT NULL,
    role VARCHAR(20) DEFAULT 'admin',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE IF NOT EXISTS students (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
```

```
    roll_no VARCHAR(30) NOT NULL UNIQUE,
    email VARCHAR(100) NOT NULL UNIQUE,
    course VARCHAR(100) NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE IF NOT EXISTS student_documents (
    id INT AUTO_INCREMENT PRIMARY KEY,
    student_id INT NOT NULL,
    file_name VARCHAR(255) NOT NULL,
    file_path VARCHAR(255) NOT NULL,
    file_type VARCHAR(100) NOT NULL,
    file_size INT NOT NULL,
    doc_category VARCHAR(50) NOT NULL,
    uploaded_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (student_id) REFERENCES students(id) ON DELETE CASCADE
);

INSERT INTO users (username, password, full_name, role)
VALUES ('admin', 'admin123', 'Administrator', 'admin')
ON DUPLICATE KEY UPDATE id=id;

INSERT INTO students (name, roll_no, email, course) VALUES
('Dhiren CP', '24CS054', 'dhiren.cp@kpriet.ac.in', 'B.E. Computer Science and Engineering'),
('Gokulnath R', '24CS021', 'gokulnath.r@kpriet.ac.in', 'B.Tech Artificial Intelligence and Data Science'),
('Sanjay Kumar M', '24CS078', 'sanjay.k@kpriet.ac.in', 'B.E. Electronics and Communication Engineering')
ON DUPLICATE KEY UPDATE id=id;
```

db.php:

```
<?php
$server = 'localhost';
$schema = 'college_portal';
$user = 'root';
$pass = '';

try {
    $db = new PDO("mysql:host=$server;dbname=$schema;charset=utf8mb4", $user, $pass, [
        PDO::ATTR_ERRMODE            => PDO::ERRMODE_EXCEPTION,
        PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
        PDO::ATTR_EMULATE_PREPARES   => false,
    ]);
} catch (PDOException $ex) {
    die("Database Connection Error: " . htmlspecialchars($ex->getMessage()));
}
?>
```

index.php:

```
<?php
require_once 'db.php';

$sql = "SELECT s.*, d.file_path AS photo_path
        FROM students s
        LEFT JOIN student_documents d
            ON s.id = d.student_id AND d.doc_category = 'profile_image'
```

```
        ORDER BY s.id DESC";
$res = $db->query($sql);
$students = $res->fetchAll();
?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Student Records - Admin Panel</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header class="top-bar-nav">
    <h1>KPRIET - Admin Dashboard</h1>
    <nav class="nav-cluster">
      <a href="index.php">Directory</a>
      <a href="add_student.php">+ New Student</a>
    </nav>
  </header>

  <main class="container-box">
    <div class="dash-heading">
      <h2>Student Records Management</h2>
      <a href="add_student.php" class="action-btn btn-create">+ Add Student Record</a>
    </div>

    <?php if (isset($_GET['msg'])): ?>
      <div class="flash-msg flash-ok"><?= htmlspecialchars($_GET['msg']) ?></div>
    <?php endif; ?>
    <?php if (isset($_GET['err'])): ?>
      <div class="flash-msg flash-err"><?= htmlspecialchars($_GET['err']) ?></div>
    <?php endif; ?>

    <div class="panel-card">
      <h3>Active Student Database (<?= count($students) ?>)</h3>
      <?php if (empty($students)): ?>
        <p>No student records found in database.</p>
      <?php else: ?>
        <table>
          <thead>
            <tr>
              <th>Profile</th>
              <th>Roll Number</th>
              <th>Student Name</th>
              <th>Degree Stream</th>
              <th>Email</th>
              <th>Actions</th>
            </tr>
          </thead>
          <tbody>
            <?php foreach ($students as $st): ?>
              <tr>
                <td>
                  <?php if (!empty($st['photo_path']) && file_exists($st['photo_path'])): ?>
```

```
                <img src="<?= htmlspecialchars($st['photo_path']) ?>" alt="Photo" class="avatar-thumbnail">
              <?php else: ?>
                <div class="avatar-none"><?= strtoupper(substr($st['name'], 0, 1)) ?></div>
              <?php endif; ?>
            </td>
            <td><strong><?= htmlspecialchars($st['roll_no']) ?></strong></td>
            <td><?= htmlspecialchars($st['name']) ?></td>
            <td><span class="badge-programme"><?= htmlspecialchars($st['course']) ?></span></td>
            <td><?= htmlspecialchars($st['email']) ?></td>
            <td>
              <div class="actions-cell">
                <a href="view_student.php?id=<?= $st['id'] ?>" class="action-btn btn-view">View</a>
                <a href="edit_student.php?id=<?= $st['id'] ?>" class="action-btn btn-edit">Edit</a>
                <a href="delete_student.php?id=<?= $st['id'] ?>" class="action-btn btn-delete" onclick="return confirm('Confirm
deletion of <?= htmlspecialchars($st['name']) ?>?');">Delete</a>
              </div>
            </td>
          </tr>
        <?php endforeach; ?>
      </tbody>
    </table>
  <?php endif; ?>
 </div>
</main>
</body>
</html>
```

add_student.php:

```
<?php
require_once 'db.php';

$errorList = [];
$uploadDir = __DIR__ . '/uploads/';
if (!is_dir($uploadDir)) {
  mkdir($uploadDir, 0777, true);
}

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
  $name = trim($_POST['name'] ?? '');
  $roll_no = trim($_POST['roll_no'] ?? '');
  $email = trim($_POST['email'] ?? '');
  $course = trim($_POST['course'] ?? '');

  if (empty($name) || empty($roll_no) || empty($email) || empty($course)) {
    $errorList[] = "All required candidate fields must be provided.";
  } elseif (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
    $errorList[] = "Please provide an authentic email format.";
  }

  if (empty($errorList)) {
    $check = $db->prepare("SELECT id FROM students WHERE roll_no = ? OR email = ?");
    $check->execute([$roll_no, $email]);
    if ($check->fetch()) {
      $errorList[] = "Roll number or email already in use.";
```

```
        }
    }

    function processUploadFile($field, $typeCategory, $allowedTypes, $maxBytes, $folder) {
        if (!isset($_FILES[$field]) || $_FILES[$field]['error'] === UPLOAD_ERR_NO_FILE) {
            return null;
        }

        $f = $_FILES[$field];
        if ($f['error'] !== UPLOAD_ERR_OK) {
            throw new Exception("File upload failed (Code: " . $f['error'] . ")");
        }

        if ($f['size'] > $maxBytes) {
            throw new Exception("File for $typeCategory exceeds max " . round($maxBytes / (1024 * 1024)) . " MB limit.");
        }

        $finfo = new finfo(FILEINFO_MIME_TYPE);
        $mime = $finfo->file($f['tmp_name']);
        if (!in_array($mime, $allowedTypes)) {
            throw new Exception("Unsupported file format ($mime) for $typeCategory.");
        }

        $ext = strtolower(pathinfo($f['name'], PATHINFO_EXTENSION));
        $uniqueFilename = $typeCategory . '_' . sha1(uniqid(mt_rand(), true)) . '.' . $ext;
        $destPath = $folder . $uniqueFilename;

        if (!move_uploaded_file($f['tmp_name'], $destPath)) {
            throw new Exception("Failed to write uploaded file to filesystem.");
        }

        return [
            'file_name' => $f['name'],
            'file_path' => 'uploads/' . $uniqueFilename,
            'file_type' => $mime,
            'file_size' => $f['size'],
            'doc_category' => $typeCategory
        ];
    }

    $uploadedFiles = [];
    if (empty($errorList)) {
        try {
            if (isset($_FILES['profile_image']) && $_FILES['profile_image']['error'] !== UPLOAD_ERR_NO_FILE) {
                $photo = processUploadFile('profile_image', 'profile_image', ['image/jpeg', 'image/png'], 2 * 1024 * 1024, $uploadDir);
                if ($photo) $uploadedFiles[] = $photo;
            }

            if (isset($_FILES['document']) && $_FILES['document']['error'] !== UPLOAD_ERR_NO_FILE) {
                $doc = processUploadFile('document', 'certificate', ['application/pdf', 'image/jpeg', 'image/png'], 5 * 1024 * 1024,
$uploadDir);
                if ($doc) $uploadedFiles[] = $doc;
            }
```

```
        } catch (Exception $e) {
            $errorList[] = $e->getMessage();
        }
    }

    if (empty($errorList)) {
        try {
            $db->beginTransaction();

            $ins = $db->prepare("INSERT INTO students (name, roll_no, email, course) VALUES (?, ?, ?, ?)");
            $ins->execute([$name, $roll_no, $email, $course]);
            $studentId = $db->lastInsertId();

            $docIns = $db->prepare("INSERT INTO student_documents (student_id, file_name, file_path, file_type, file_size,
doc_category) VALUES (?, ?, ?, ?, ?, ?)");
            foreach ($uploadedFiles as $docMeta) {
                $docIns->execute([
                    $studentId,
                    $docMeta['file_name'],
                    $docMeta['file_path'],
                    $docMeta['file_type'],
                    $docMeta['file_size'],
                    $docMeta['doc_category']
                ]);
            }

            $db->commit();
            header("Location: index.php?msg=" . urlencode("Student record and files committed successfully."));
            exit;
        } catch (PDOException $e) {
            $db->rollBack();
            $errorList[] = "Database Error: " . $e->getMessage();
        }
    }
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Add Student - Admin Portal</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header class="top-bar-nav">
    <h1>KPRIET - Admin Dashboard</h1>
    <nav class="nav-cluster"><a href="index.php">&larr; Back to Directory</a></nav>
  </header>

  <main class="container-box">
    <div class="panel-card">
      <h3>Register Student Record &amp;amp; File Attachments</h3>

      <?php if (!empty($errorList)): ?>
```

```
      <div class="flash-msg flash-err">
        <ul><?php foreach ($errorList as $err) echo "<li>" . htmlspecialchars($err) . "</li>"; ?></ul>
      </div>
    <?php endif; ?>

    <form action="add_student.php" method="POST" enctype="multipart/form-data">
      <div class="field-group">
        <label>Full Name *</label>
        <input type="text" name="name" required value="<?= htmlspecialchars($_POST['name'] ?? '') ?>" placeholder="e.g. Dhiren
CP">
      </div>
      <div class="field-group">
        <label>Roll Number *</label>
        <input type="text" name="roll_no" required value="<?= htmlspecialchars($_POST['roll_no'] ?? '') ?>" placeholder="e.g.
24CS054">
      </div>
      <div class="field-group">
        <label>Email Address *</label>
        <input type="email" name="email" required value="<?= htmlspecialchars($_POST['email'] ?? '') ?>" placeholder="e.g.
dhiren.cp@kpriet.ac.in">
      </div>
      <div class="field-group">
        <label>Programme *</label>
        <select name="course" required>
          <option value="">-- Choose Programme --</option>
          <option value="B.E. Computer Science and Engineering">B.E. Computer Science and Engineering</option>
          <option value="B.Tech Artificial Intelligence and Data Science">B.Tech Artificial Intelligence and Data Science</option>
          <option value="B.Tech Information Technology">B.Tech Information Technology</option>
          <option value="B.E. Electronics and Communication Engineering">B.E. Electronics and Communication
Engineering</option>
        </select>
      </div>
      <div class="field-group">
        <label>Profile Picture (JPG/PNG, Max 2MB)</label>
        <input type="file" name="profile_image" accept="image/jpeg,image/png">
      </div>
      <div class="field-group">
        <label>ID Proof / Academic Certificate (PDF/JPG/PNG, Max 5MB)</label>
        <input type="file" name="document" accept="application/pdf,image/jpeg,image/png">
      </div>
      <div style="margin-top: 20px;">
        <button type="submit" class="action-btn btn-create">Save Record</button>
        <a href="index.php" class="action-btn btn-back">Cancel</a>
      </div>
    </form>
  </div>
 </main>
</body>
</html>
```

view_student.php:

```
<?php
require_once 'db.php';

$studentId = $_GET['id'] ?? null;
```

```
if (!$studentId || !is_numeric($studentId)) {
    header("Location: index.php?err=" . urlencode("Invalid identifier"));
    exit;
}

$stmt = $db->prepare("SELECT * FROM students WHERE id = ?");
$stmt->execute([$studentId]);
$student = $stmt->fetch();

if (!$student) {
    header("Location: index.php?err=" . urlencode("Student record not located"));
    exit;
}

$docQuery = $db->prepare("SELECT * FROM student_documents WHERE student_id = ? ORDER BY uploaded_at DESC");
$docQuery->execute([$studentId]);
$documents = $docQuery->fetchAll();

$profilePhoto = null;
foreach ($documents as $doc) {
    if ($doc['doc_category'] === 'profile_image' && file_exists($doc['file_path'])) {
        $profilePhoto = $doc['file_path'];
        break;
    }
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>View Profile - <?= htmlspecialchars($student['name']) ?></title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header class="top-bar-nav">
    <h1>KPRIET - Admin Dashboard</h1>
    <nav class="nav-cluster">
      <a href="index.php">&larr; Back to Directory</a>
      <a href="edit_student.php?id=<?= $student['id'] ?>">Edit</a>
    </nav>
  </header>

  <main class="container-box">
    <div class="panel-card">
      <div class="profile-card-inner">
        <div>
          <?php if ($profilePhoto): ?>
            <img src="<?= htmlspecialchars($profilePhoto) ?>" alt="Photo" class="large-photo">
          <?php else: ?>
            <div class="large-photo" style="background:#e2e8f0;display:flex;align-items:center;justify-content:center;font-
size:36px;color:#64748b;">
              <?= strtoupper(substr($student['name'], 0, 1)) ?>
            </div>
          <?php endif; ?>
```

```
      </div>
      <div class="profile-text">
        <h3><?= htmlspecialchars($student['name']) ?></h3>
        <p><strong>Roll Number:</strong> <?= htmlspecialchars($student['roll_no']) ?></p>
        <p><strong>Email Address:</strong> <?= htmlspecialchars($student['email']) ?></p>
        <p><strong>Degree Programme:</strong> <span class="badge-programme"><?= htmlspecialchars($student['course'])
?></span></p>
      </div>
    </div>
  </div>

  <div class="panel-card">
    <h3>Uploaded Documents &amp;amp; Metadata (<?= count($documents) ?>)</h3>
    <?php if (empty($documents)): ?>
      <p>No attached files registered for this candidate.</p>
    <?php else: ?>
      <table>
        <thead>
          <tr>
            <th>Category</th>
            <th>Original Filename</th>
            <th>MIME Type</th>
            <th>File Size</th>
            <th>Action</th>
          </tr>
        </thead>
        <tbody>
          <?php foreach ($documents as $d): ?>
            <tr>
              <td><span class="badge-programme"><?= htmlspecialchars(ucwords(str_replace('_', ' ', $d['doc_category'])))
?></span></td>
              <td><?= htmlspecialchars($d['file_name']) ?></td>
              <td><code><?= htmlspecialchars($d['file_type']) ?></code></td>
              <td><?= round($d['file_size'] / 1024, 1) ?> KB</td>
              <td>
                <a href="<?= htmlspecialchars($d['file_path']) ?>" target="_blank" class="action-btn btn-view"
download>Download</a>
              </td>
            </tr>
          <?php endforeach; ?>
        </tbody>
      </table>
    <?php endif; ?>
  </div>
 </main>
</body>
</html>
```

edit_student.php:

```
<?php
require_once 'db.php';

$studentId = $_GET['id'] ?? null;
if (!$studentId || !is_numeric($studentId)) {
```

```
    header("Location: index.php");
    exit;
}

$findStmt = $db->prepare("SELECT * FROM students WHERE id = ?");
$findStmt->execute([$studentId]);
$student = $findStmt->fetch();

if (!$student) {
    header("Location: index.php");
    exit;
}

$errors = [];
$dest = __DIR__ . '/uploads/';

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $name = trim($_POST['name'] ?? '');
    $roll_no = trim($_POST['roll_no'] ?? '');
    $email = trim($_POST['email'] ?? '');
    $course = trim($_POST['course'] ?? '');

    if (empty($name) || empty($roll_no) || empty($email) || empty($course)) {
        $errors[] = "All required fields must be supplied.";
    }

    if (empty($errors) && isset($_FILES['profile_image']) && $_FILES['profile_image']['error'] !== UPLOAD_ERR_NO_FILE) {
        $file = $_FILES['profile_image'];
        $finfo = new finfo(FILEINFO_MIME_TYPE);
        $mime = $finfo->file($file['tmp_name']);

        if (in_array($mime, ['image/jpeg', 'image/png']) && $file['size'] <= 2 * 1024 * 1024) {
            $priorStmt = $db->prepare("SELECT id, file_path FROM student_documents WHERE student_id = ? AND doc_category =
'profile_image'");
            $priorStmt->execute([$studentId]);
            $priorDoc = $priorStmt->fetch();
            if ($priorDoc && file_exists($priorDoc['file_path'])) {
                @unlink($priorDoc['file_path']);
            }

            $ext = strtolower(pathinfo($file['name'], PATHINFO_EXTENSION));
            $newName = 'profile_' . sha1(uniqid(mt_rand(), true)) . '.' . $ext;
            if (move_uploaded_file($file['tmp_name'], $dest . $newName)) {
                if ($priorDoc) {
                    $upDoc = $db->prepare("UPDATE student_documents SET file_name=?, file_path=?, file_type=?, file_size=?
WHERE id=?");
                    $upDoc->execute([$file['name'], 'uploads/' . $newName, $mime, $file['size'], $priorDoc['id']]);
                } else {
                    $inDoc = $db->prepare("INSERT INTO student_documents (student_id, file_name, file_path, file_type, file_size,
doc_category) VALUES (?, ?, ?, ?, ?, 'profile_image')");
                    $inDoc->execute([$studentId, $file['name'], 'uploads/' . $newName, $mime, $file['size']]);
                }
            }
```

```
        }
    }

    if (empty($errors)) {
        $upStudent = $db->prepare("UPDATE students SET name = ?, roll_no = ?, email = ?, course = ? WHERE id = ?");
        $upStudent->execute([$name, $roll_no, $email, $course, $studentId]);
        header("Location: view_student.php?id=$studentId&msg=" . urlencode("Student updated successfully"));
        exit;
    }
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Edit Record - <?= htmlspecialchars($student['name']) ?></title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header class="top-bar-nav">
    <h1>KPRIET - Admin Dashboard</h1>
    <nav class="nav-cluster"><a href="index.php">&larr; Back to Directory</a></nav>
  </header>
  <main class="container-box">
    <div class="panel-card">
      <h3>Edit Student Details</h3>
      <form action="edit_student.php?id=<?= $student['id'] ?>" method="POST" enctype="multipart/form-data">
        <div class="field-group">
          <label>Full Name</label>
          <input type="text" name="name" value="<?= htmlspecialchars($student['name']) ?>" required>
        </div>
        <div class="field-group">
          <label>Roll Number</label>
          <input type="text" name="roll_no" value="<?= htmlspecialchars($student['roll_no']) ?>" required>
        </div>
        <div class="field-group">
          <label>Email Address</label>
          <input type="email" name="email" value="<?= htmlspecialchars($student['email']) ?>" required>
        </div>
        <div class="field-group">
          <label>Programme</label>
          <input type="text" name="course" value="<?= htmlspecialchars($student['course']) ?>" required>
        </div>
        <div class="field-group">
          <label>Replace Photo (Optional)</label>
          <input type="file" name="profile_image" accept="image/jpeg,image/png">
        </div>
        <button type="submit" class="action-btn btn-edit">Save Updates</button>
      </form>
    </div>
  </main>
</body>
</html>
```

delete_student.php:

```
<?php
require_once 'db.php';

$studentId = $_GET['id'] ?? null;
if ($studentId && is_numeric($studentId)) {
    $fetchFiles = $db->prepare("SELECT file_path FROM student_documents WHERE student_id = ?");
    $fetchFiles->execute([$studentId]);
    $files = $fetchFiles->fetchAll();

    foreach ($files as $f) {
        $abs = __DIR__ . '/' . $f['file_path'];
        if (file_exists($abs) && is_file($abs)) {
            @unlink($abs);
        }
    }

    $delStmt = $db->prepare("DELETE FROM students WHERE id = ?");
    $delStmt->execute([$studentId]);

    header("Location: index.php?msg=" . urlencode("Student and uploaded files deleted successfully."));
    exit;
}
header("Location: index.php");
?>
```

style.css:

```
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, Arial, sans-serif;
  background-color: #f8fafc;
  color: #0f172a;
  line-height: 1.5;
}

.top-bar-nav {
  background-color: #312e81;
  color: #ffffff;
  padding: 16px 5%;
  display: flex;
  justify-content: space-between;
  align-items: center;
  border-bottom: 3px solid #06b6d4;
}

.top-bar-nav h1 {
  font-size: 20px;
}
```

```
.nav-cluster a {
  color: #c7d2fe;
  text-decoration: none;
  font-weight: 600;
  margin-left: 18px;
  font-size: 13.5px;
}

.nav-cluster a:hover {
  color: #ffffff;
}

.container-box {
  width: 94%;
  max-width: 1080px;
  margin: 28px auto;
}

.dash-heading {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 18px;
  flex-wrap: wrap;
  gap: 12px;
}

.panel-card {
  background: #ffffff;
  border-radius: 10px;
  padding: 24px;
  border: 1px solid #e2e8f0;
  box-shadow: 0 4px 12px rgba(0, 0, 0, 0.03);
  margin-bottom: 24px;
}

.panel-card h3 {
  color: #312e81;
  border-left: 3px solid #4f46e5;
  padding-left: 10px;
  margin-bottom: 16px;
  font-size: 16px;
}

table {
  width: 100%;
  border-collapse: collapse;
}

th, td {
  padding: 11px 13px;
  text-align: left;
```

```
  border-bottom: 1px solid #f1f5f9;
  font-size: 13.5px;
}

th {
  background-color: #312e81;
  color: #ffffff;
  font-weight: 600;
}

tr:hover {
  background-color: #f8fafc;
}

.avatar-thumbnail {
  width: 44px;
  height: 44px;
  border-radius: 50%;
  object-fit: cover;
  border: 2px solid #cbd5e1;
}

.avatar-none {
  width: 44px;
  height: 44px;
  border-radius: 50%;
  background-color: #e2e8f0;
  display: flex;
  align-items: center;
  justify-content: center;
  font-weight: bold;
  color: #64748b;
  font-size: 14px;
}

.action-btn {
  display: inline-block;
  padding: 7px 14px;
  font-size: 12.5px;
  font-weight: 600;
  border-radius: 5px;
  text-decoration: none;
  cursor: pointer;
  border: none;
}

.btn-create { background: #4f46e5; color: #ffffff; }
.btn-view { background: #0284c7; color: #ffffff; }
.btn-edit { background: #d97706; color: #ffffff; }
.btn-delete { background: #e11d48; color: #ffffff; }
.btn-back { background: #64748b; color: #ffffff; }

.actions-cell { display: flex; gap: 6px; }
```

```
.badge-programme {
  display: inline-block;
  padding: 3px 8px;
  border-radius: 4px;
  font-size: 11px;
  font-weight: 600;
  background: #ede9fe;
  color: #5b21b6;
}

.flash-msg {
  padding: 12px 16px;
  border-radius: 6px;
  margin-bottom: 18px;
  font-size: 13.5px;
  font-weight: 500;
}
.flash-ok { background: #ecfdf5; color: #065f46; border: 1px solid #a7f3d0; }
.flash-err { background: #fff1f2; color: #9f1239; border: 1px solid #fecdd3; }

.field-group {
  margin-bottom: 14px;
}

.field-group label {
  display: block;
  font-weight: 600;
  font-size: 13px;
  margin-bottom: 5px;
  color: #334155;
}

.field-group input,
.field-group select {
  width: 100%;
  padding: 9px 12px;
  border: 1px solid #cbd5e1;
  border-radius: 6px;
  font-size: 13.5px;
}

.profile-card-inner {
  display: flex;
  gap: 24px;
  align-items: center;
}

.large-photo {
  width: 120px;
  height: 120px;
  border-radius: 8px;
  object-fit: cover;
```

```
  border: 2px solid #4f46e5;
}

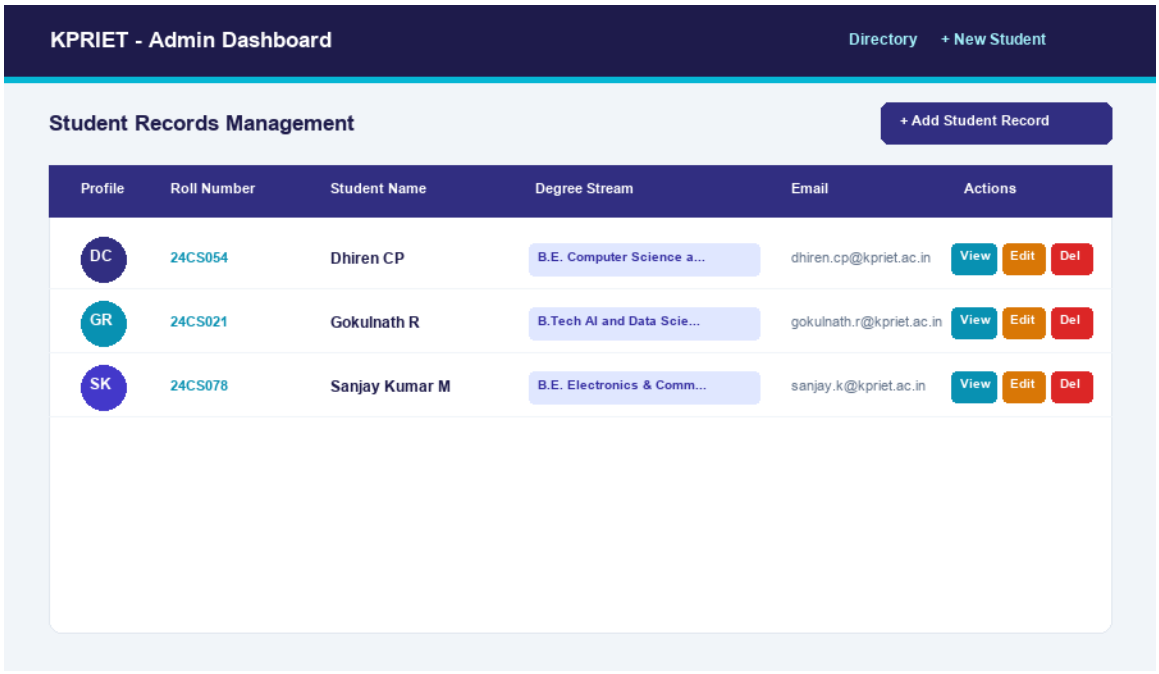
.profile-text p {
  margin-bottom: 6px;
  font-size: 14px;
}

.profile-text strong {
  min-width: 110px;
  display: inline-block;
  color: #312e81;
}
```

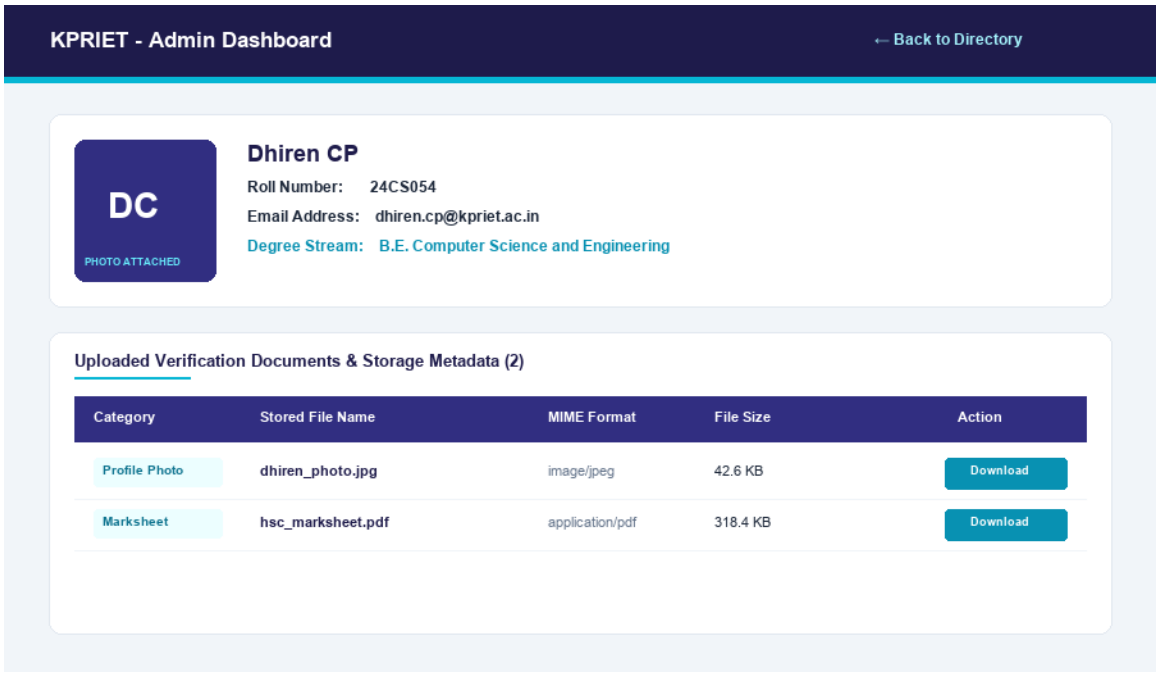
OUTPUT :

VALID OPERATIONS :

Valid Operation – Student Records Directory & Action Management (CRUD Read) :



Valid Operation – Detailed Dossier with Uploaded Files Metadata :



RUBRICS:

Rubrics	Actual Marks	Awarded Marks

Relational Database Schema & Integrity (R1)	4	
Complete CRUD Implementation (R2)	4	
File Upload & MIME Verification (R3)	4	
Storage Security & File Unlink Management (R4)	4	
UI Presentation & Flash Messaging (R5)	4	
Total Marks	**20**	

RESULT:

A full-stack College Administration Panel for student records and file attachments was successfully implemented using PHP and MySQL. The system executes parameterized CRUD transactions, strictly verifies file size and MIME properties, assigns hashed collision-free names, tracks comprehensive document metadata, and maintains synchronized physical storage by unlinking files during record deletions.

Q No: 3	Dynamic student profiles portal with image & document streaming using Java Servlets, JDBC & JSP

AIM:

To develop an enterprise Java Web Application using Java Servlets, JDBC, and JavaServer Pages (JSP) to render dynamic student profile cards, retrieve and byte-stream images and documents from server storage using <img> tags pointing directly to servlet endpoints, apply conditional JSP rendering for document attachments, and enforce session-based administrative access control.

1. Configure web.xml deployment descriptor with declarations and URL mappings for AuthenticationServlet (/auth), PortalLogoutServlet (/portalLogout), StudentProfileServlet (/profiles), and AssetStreamServlet (/streamAsset).
2. Set standard session timeout to 30 minutes in web.xml.
3. Implement DBConfig.java to load the MySQL Connector/J driver (com.mysql.cj.jdbc.Driver) and manage relational JDBC connections.
4. Develop AuthenticationServlet (doPost) to authenticate administrative credentials against the MySQL users table and establish an active HttpSession.
5. Develop PortalLogoutServlet (doGet) to invalidate active sessions and redirect back to login.jsp.
6. Develop StudentProfileServlet (doGet) to verify user authentication, query student profiles and associated document keys via SQL JOIN, and forward the result set to students.jsp.
7. Develop AssetStreamServlet (doGet) to read the document ID parameter, fetch storage metadata from MySQL, locate physical files on the server filesystem, set MIME Content-Type and Content-Disposition headers, and pipe raw bytes to ServletOutputStream using buffered reading.
8. Design login.jsp providing a responsive administrative sign-in portal styled with the Royal Indigo theme.
9. Design students.jsp with session verification to restrict unauthenticated access.
10. Utilize dynamic JSP scriptlets (<% for (...) %>) to iterate across profile records and render responsive profile cards.
11. Embed <img> tags referencing servlet streaming endpoints (<img src='streamAsset?id=...'>) to display student photographs dynamically.
12. Incorporate conditional rendering logic to display document download links when present or an inactive tag when no files are linked.
13. Test administrative sign-in, dynamic profile card rendering, servlet binary streaming, document downloads, and session invalidation.

SERVLET & JSP CONSTRUCTS USED:

web.xml
<servlet>
<servlet-name>
<servlet-class>
<servlet-mapping>
<url-pattern>
<session-config>
<welcome-file-list>
HttpServlet

doGet()
doPost()
HttpServletRequest
HttpServletResponse
HttpSession
request.getSession()
session.setAttribute()
session.getAttribute()
session.invalidate()
RequestDispatcher
dispatcher.forward()
response.sendRedirect()
response.setContentType()
response.setContentLengthLong()
response.setHeader()
response.getOutputStream()
FileInputStream
File
<%@ page %>
<% %>
<%= %>

JDBC CONSTRUCTS USED:

Class.forName("com.mysql.cj.jdbc.Driver")
DriverManager.getConnection()
Connection
PreparedStatement
ResultSet
executeQuery()
prepareStatement()
conn.close()
stmt.close()
rs.close()
SQLException

web.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee
                        http://xmlns.jcp.org/xml/ns/javaee/web-app_3_1.xsd"
        version="3.1">

    <display-name>KPRIET Verified Student Profile Portal</display-name>

    <welcome-file-list>
        <welcome-file>login.jsp</welcome-file>
    </welcome-file-list>

    <servlet>
        <servlet-name>AuthenticationServlet</servlet-name>
```

```
    <servlet-class>AuthenticationServlet</servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name>AuthenticationServlet</servlet-name>
    <url-pattern>/auth</url-pattern>
  </servlet-mapping>

  <servlet>
    <servlet-name>PortalLogoutServlet</servlet-name>
    <servlet-class>PortalLogoutServlet</servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name>PortalLogoutServlet</servlet-name>
    <url-pattern>/portalLogout</url-pattern>
  </servlet-mapping>

  <servlet>
    <servlet-name>StudentProfileServlet</servlet-name>
    <servlet-class>StudentProfileServlet</servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name>StudentProfileServlet</servlet-name>
    <url-pattern>/profiles</url-pattern>
  </servlet-mapping>

  <servlet>
    <servlet-name>AssetStreamServlet</servlet-name>
    <servlet-class>AssetStreamServlet</servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name>AssetStreamServlet</servlet-name>
    <url-pattern>/streamAsset</url-pattern>
  </servlet-mapping>

  <session-config>
    <session-timeout>30</session-timeout>
  </session-config>
</web-app>
```

DBConfig.java:

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class DBConfig {
  private static final String DB_URL =
"jdbc:mysql://localhost:3306/college_portal?useSSL=false&allowPublicKeyRetrieval=true&serverTimezone=UTC";
  private static final String DB_USER = "root";
  private static final String DB_PASS = "";

  static {
    try {
      Class.forName("com.mysql.cj.jdbc.Driver");
    } catch (ClassNotFoundException e) {
```

```
            e.printStackTrace();
        }
    }

    public static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(DB_URL, DB_USER, DB_PASS);
    }
}
```

AuthenticationServlet.java:

```
import java.io.IOException;
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

public class AuthenticationServlet extends HttpServlet {

    @Override
    protected void doPost(HttpServletRequest req, HttpServletResponse resp)
            throws ServletException, IOException {

        String userParam = req.getParameter("username");
        String passParam = req.getParameter("password");

        if (userParam == null || passParam == null || userParam.trim().isEmpty() || passParam.trim().isEmpty()) {
            resp.sendRedirect("login.jsp?error=Username+and+Password+required");
            return;
        }

        try (Connection connection = DBConfig.getConnection();
             PreparedStatement statement = connection.prepareStatement("SELECT * FROM users WHERE username = ? AND
password = ?")) {

            statement.setString(1, userParam.trim());
            statement.setString(2, passParam.trim());
            ResultSet rs = statement.executeQuery();

            if (rs.next()) {
                HttpSession activeSession = req.getSession(true);
                activeSession.setAttribute("authenticatedUser", rs.getString("username"));
                activeSession.setAttribute("userDisplayName", rs.getString("full_name"));
                activeSession.setAttribute("userRole", rs.getString("role"));
                resp.sendRedirect("profiles");
            } else {
                resp.sendRedirect("login.jsp?error=Invalid+administrative+credentials");
            }
        } catch (Exception ex) {
            resp.sendRedirect("login.jsp?error=" + ex.getMessage());
```

```
        }
    }
}
```

PortalLogoutServlet.java:

```
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

public class PortalLogoutServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse resp)
            throws ServletException, IOException {

        HttpSession existingSession = req.getSession(false);
        if (existingSession != null) {
            existingSession.invalidate();
        }
        resp.sendRedirect("login.jsp?msg=Successfully+logged+out+from+portal");
    }
}
```

StudentProfileServlet.java:

```
import java.io.IOException;
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.ArrayList;
import java.util.HashMap;
import java.util.List;
import java.util.Map;
import javax.servlet.RequestDispatcher;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

public class StudentProfileServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse resp)
            throws ServletException, IOException {

        HttpSession currentSession = req.getSession(false);
        if (currentSession == null || currentSession.getAttribute("authenticatedUser") == null) {
            resp.sendRedirect("login.jsp?error=Authentication+required+to+access+profiles");
            return;
        }
```

```
        List<Map<String, Object>> profileRecords = new ArrayList<>();

        try (Connection conn = DBConfig.getConnection()) {
            String query = "SELECT s.id AS student_pk, s.name, s.roll_no, s.email, s.course, " +
                           "       d1.id AS photo_doc_id, d2.id AS cert_doc_id " +
                           "FROM students s " +
                           "LEFT JOIN student_documents d1 ON s.id = d1.student_id AND d1.doc_category = 'profile_image' " +
                           "LEFT JOIN student_documents d2 ON s.id = d2.student_id AND d2.doc_category != 'profile_image' " +
                           "ORDER BY s.id ASC";

            PreparedStatement stmt = conn.prepareStatement(query);
            ResultSet rs = stmt.executeQuery();

            while (rs.next()) {
                Map<String, Object> item = new HashMap<>();
                item.put("id", rs.getInt("student_pk"));
                item.put("name", rs.getString("name"));
                item.put("roll_no", rs.getString("roll_no"));
                item.put("email", rs.getString("email"));
                item.put("course", rs.getString("course"));
                item.put("photo_id", rs.getObject("photo_doc_id"));
                item.put("doc_id", rs.getObject("cert_doc_id"));
                profileRecords.add(item);
            }

            req.setAttribute("profileRecords", profileRecords);
            RequestDispatcher viewDispatcher = req.getRequestDispatcher("students.jsp");
            viewDispatcher.forward(req, resp);

        } catch (Exception err) {
            resp.sendError(HttpServletResponse.SC_INTERNAL_SERVER_ERROR, "Database retrieval error: " + err.getMessage());
        }
    }
}
```

AssetStreamServlet.java:

```
import java.io.File;
import java.io.FileInputStream;
import java.io.IOException;
import java.io.OutputStream;
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

public class AssetStreamServlet extends HttpServlet {

    private static final int CHUNK_SIZE = 8192;
```

```
    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse resp)
            throws ServletException, IOException {

        HttpSession sess = req.getSession(false);
        if (sess == null || sess.getAttribute("authenticatedUser") == null) {
            resp.sendError(HttpServletResponse.SC_UNAUTHORIZED, "Unauthorized access rejected");
            return;
        }

        String rawDocId = req.getParameter("id");
        if (rawDocId == null || rawDocId.trim().isEmpty()) {
            resp.sendError(HttpServletResponse.SC_BAD_REQUEST, "Required parameter id is missing");
            return;
        }

        try (Connection conn = DBConfig.getConnection();
             PreparedStatement ps = conn.prepareStatement("SELECT file_name, file_path, file_type FROM student_documents
WHERE id = ?")) {

            ps.setInt(1, Integer.parseInt(rawDocId.trim()));
            ResultSet rs = ps.executeQuery();

            if (!rs.next()) {
                resp.sendError(HttpServletResponse.SC_NOT_FOUND, "Document record not found");
                return;
            }

            String storedFileName = rs.getString("file_name");
            String storedFilePath = rs.getString("file_path");
            String mimeType = rs.getString("file_type");

            File targetFile = new File(getServletContext().getRealPath("/") + storedFilePath);
            if (!targetFile.exists()) {
                targetFile = new File(storedFilePath);
            }

            if (!targetFile.exists()) {
                resp.sendError(HttpServletResponse.SC_NOT_FOUND, "Physical file unavailable on storage");
                return;
            }

            resp.setContentType(mimeType != null ? mimeType : "application/octet-stream");
            resp.setContentLengthLong(targetFile.length());

            boolean isPreviewable = mimeType != null && (mimeType.startsWith("image/") || mimeType.equals("application/pdf"));
            resp.setHeader("Content-Disposition", isPreviewable ? "inline" : "inline; filename=\"" + storedFileName + "\"");

            try (FileInputStream fileIn = new FileInputStream(targetFile);
                 OutputStream streamOut = resp.getOutputStream()) {

                byte[] buffer = new byte[CHUNK_SIZE];
                int readLength;
```

```
            while ((readLength = fileIn.read(buffer)) != -1) {
                streamOut.write(buffer, 0, readLength);
            }
            streamOut.flush();
        }

    } catch (Exception ex) {
        resp.sendError(HttpServletResponse.SC_INTERNAL_SERVER_ERROR, "Asset stream error: " + ex.getMessage());
    }
  }
}
```

login.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Admin Portal Sign-In - KPRIET</title>
  <link rel="stylesheet" href="style.css">
</head>
<body class="auth-viewport">

  <div class="auth-card">
    <div class="auth-header">
      <div class="auth-logo-badge">KPR</div>
      <h2>Student Profiles Portal</h2>
      <p>Administrative Verification Gateway</p>
    </div>

    <%
      String error = request.getParameter("error");
      if (error != null && !error.trim().isEmpty()) {
    %>
      <div class="notice-banner notice-danger"><%= error %></div>
    <% } %>

    <%
      String msg = request.getParameter("msg");
      if (msg != null && !msg.trim().isEmpty()) {
    %>
      <div class="notice-banner notice-success"><%= msg %></div>
    <% } %>

    <form action="auth" method="POST" class="auth-form-body">
      <div class="field-group">
        <label for="username">Administrative Username</label>
        <input type="text" id="username" name="username" placeholder="e.g. admin" required autofocus>
      </div>

      <div class="field-group">
        <label for="password">Security Password</label>
        <input type="password" id="password" name="password" placeholder="e.g. admin123" required>
```

```
    </div>

    <button type="submit" class="auth-submit-btn">Authenticate &amp;amp; Enter</button>
   </form>

   <div class="auth-card-footer">
    <span>KPR Institute of Engineering and Technology</span>
   </div>
  </div>

</body>
</html>
```

students.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>
<%@ page import="java.util.List, java.util.Map" %>
<%
   if (session == null || session.getAttribute("authenticatedUser") == null) {
      response.sendRedirect("login.jsp?error=Restricted+area.+Please+authenticate.");
      return;
   }

   String activeAdmin = (String) session.getAttribute("userDisplayName");
   if (activeAdmin == null) {
      activeAdmin = (String) session.getAttribute("authenticatedUser");
   }
   List<Map<String, Object>> profileRecords = (List<Map<String, Object>>) request.getAttribute("profileRecords");
%>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Dynamic Student Profiles - KPRIET</title>
  <link rel="stylesheet" href="style.css">
</head>
<body class="portal-body">

  <header class="masthead">
   <div class="masthead-inner">
    <div class="brand-group">
     <span class="portal-tag">KPRIET</span>
     <h1>Dynamic Student Profiles Portal</h1>
    </div>
    <div class="admin-meta-cluster">
     <span class="meta-label">Active Session: <strong><%= activeAdmin %></strong></span>
     <a href="portalLogout" class="portal-logout-link">Sign Out</a>
    </div>
   </div>
  </header>

  <main class="portal-viewport">
   <div class="directory-hero">
    <div class="hero-left">
```

```
    <h2>Verified Academic Student Records</h2>
    <p>Servlet JDBC dynamic retrieval with live binary image and document streaming</p>
   </div>
   <div class="hero-right">
    <span class="counter-badge"><%= (profileRecords != null ? profileRecords.size() : 0) %> Enrolled Students</span>
   </div>
  </div>

  <% if (profileRecords == null || profileRecords.isEmpty()) { %>
   <div class="empty-state-notice">
    <p>No verified student profiles found in database repository.</p>
   </div>
  <% } else { %>
   <div class="cards-layout-grid">
    <%
     for (Map<String, Object> item : profileRecords) {
      Object photoId = item.get("photo_id");
      Object docId = item.get("doc_id");
      String stuName = (String) item.get("name");
      String stuRoll = (String) item.get("roll_no");
      String stuCourse = (String) item.get("course");
      String stuEmail = (String) item.get("email");
    %>
     <div class="academic-card">
      <div class="avatar-banner">
       <% if (photoId != null) { %>
        <img src="streamAsset?id=<%= photoId %>" alt="<%= stuName %>" class="avatar-image">
       <% } else { %>
        <div class="avatar-letter-badge"><%= stuName.substring(0, 1).toUpperCase() %></div>
       <% } %>
      </div>
      <div class="card-details-panel">
       <span class="roll-pill"><%= stuRoll %></span>
       <h3 class="student-name-title"><%= stuName %></h3>
       <p class="course-text"><%= stuCourse %></p>
       <div class="email-row">
        <span class="email-icon">&#9993;</span>
        <span class="email-string"><%= stuEmail %></span>
       </div>
       <div class="card-attachment-panel">
        <% if (docId != null) { %>
         <a href="streamAsset?id=<%= docId %>" target="_blank" class="attachment-btn">
          View Uploaded Document
         </a>
        <% } else { %>
         <span class="no-doc-tag">No Attached File</span>
        <% } %>
       </div>
      </div>
     </div>
    <% } %>
   </div>
  <% } %>
</main>
```

```
</body>
</html>
```

style.css:

```
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, "Helvetica Neue", Arial, sans-serif;
  background-color: #f1f5f9;
  color: #1e293b;
  line-height: 1.5;
}

.portal-body {
  min-height: 100vh;
  display: flex;
  flex-direction: column;
}

.masthead {
  background: linear-gradient(135deg, #1e1b4b 0%, #312e81 60%, #4338ca 100%);
  color: #ffffff;
  padding: 16px 24px;
  box-shadow: 0 4px 14px rgba(30, 27, 75, 0.25);
  border-bottom: 2px solid #06b6d4;
}

.masthead-inner {
  max-width: 1200px;
  margin: 0 auto;
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
  gap: 12px;
}

.brand-group {
  display: flex;
  align-items: center;
  gap: 12px;
}

.portal-tag {
  background-color: #06b6d4;
  color: #082f49;
  font-size: 11px;
  font-weight: 800;
```

```
  padding: 4px 8px;
  border-radius: 4px;
  letter-spacing: 0.5px;
}

.brand-group h1 {
  font-size: 19px;
  font-weight: 700;
  letter-spacing: -0.2px;
}

.admin-meta-cluster {
  display: flex;
  align-items: center;
  gap: 16px;
  font-size: 13.5px;
}

.meta-label strong {
  color: #67e8f9;
}

.portal-logout-link {
  background: rgba(239, 68, 68, 0.9);
  color: #ffffff;
  text-decoration: none;
  padding: 6px 14px;
  border-radius: 6px;
  font-size: 12.5px;
  font-weight: 600;
  transition: background-color 0.2s;
}

.portal-logout-link:hover {
  background: #dc2626;
}

.portal-viewport {
  max-width: 1200px;
  width: 92%;
  margin: 32px auto 48px auto;
  flex: 1;
}

.directory-hero {
  display: flex;
  justify-content: space-between;
  align-items: flex-end;
  margin-bottom: 28px;
  border-bottom: 2px solid #e2e8f0;
  padding-bottom: 16px;
  flex-wrap: wrap;
  gap: 12px;
```

```
}

.hero-left h2 {
  font-size: 22px;
  font-weight: 700;
  color: #1e1b4b;
  margin-bottom: 4px;
}

.hero-left p {
  font-size: 13.5px;
  color: #64748b;
}

.counter-badge {
  background-color: #e0e7ff;
  color: #3730a3;
  padding: 6px 14px;
  border-radius: 20px;
  font-size: 12.5px;
  font-weight: 700;
  border: 1px solid #c7d2fe;
}

.empty-state-notice {
  background: #ffffff;
  border-radius: 12px;
  padding: 48px;
  text-align: center;
  color: #64748b;
  box-shadow: 0 2px 8px rgba(0,0,0,0.04);
}

.cards-layout-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(280px, 1fr));
  gap: 24px;
}

.academic-card {
  background: #ffffff;
  border-radius: 14px;
  overflow: hidden;
  border: 1px solid #e2e8f0;
  box-shadow: 0 4px 12px rgba(15, 23, 42, 0.05);
  display: flex;
  flex-direction: column;
  transition: transform 0.2s, box-shadow 0.2s;
}

.academic-card:hover {
  transform: translateY(-3px);
  box-shadow: 0 8px 20px rgba(49, 46, 129, 0.12);
```

```
}

.avatar-banner {
  background: linear-gradient(135deg, #312e81 0%, #4338ca 100%);
  padding: 24px 0;
  display: flex;
  justify-content: center;
  align-items: center;
  border-bottom: 3px solid #06b6d4;
}

.avatar-image {
  width: 94px;
  height: 94px;
  border-radius: 50%;
  object-fit: cover;
  border: 3px solid #ffffff;
  box-shadow: 0 4px 10px rgba(0,0,0,0.25);
}

.avatar-letter-badge {
  width: 94px;
  height: 94px;
  border-radius: 50%;
  background-color: #e0e7ff;
  color: #312e81;
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 36px;
  font-weight: 800;
  border: 3px solid #ffffff;
  box-shadow: 0 4px 10px rgba(0,0,0,0.25);
}

.card-details-panel {
  padding: 20px 18px;
  flex: 1;
  display: flex;
  flex-direction: column;
}

.roll-pill {
  display: inline-block;
  background-color: #ecfeff;
  color: #0e7490;
  border: 1px solid #a5f3fc;
  font-size: 11px;
  font-weight: 800;
  padding: 3px 8px;
  border-radius: 6px;
  align-self: flex-start;
  margin-bottom: 10px;
  letter-spacing: 0.5px;
```

```
}

.student-name-title {
  font-size: 17px;
  font-weight: 700;
  color: #1e1b4b;
  margin-bottom: 6px;
}

.course-text {
  font-size: 12.5px;
  color: #475569;
  line-height: 1.4;
  margin-bottom: 10px;
}

.email-row {
  display: flex;
  align-items: center;
  gap: 8px;
  font-size: 12px;
  color: #64748b;
  margin-bottom: 18px;
}

.email-icon {
  color: #4f46e5;
  font-size: 14px;
}

.card-attachment-panel {
  margin-top: auto;
  border-top: 1px solid #f1f5f9;
  padding-top: 14px;
}

.attachment-btn {
  display: block;
  text-align: center;
  background-color: #f8fafc;
  color: #312e81;
  border: 1px solid #cbd5e1;
  padding: 9px 12px;
  border-radius: 8px;
  text-decoration: none;
  font-size: 12.5px;
  font-weight: 700;
  transition: all 0.2s;
}

.attachment-btn:hover {
  background-color: #312e81;
  color: #ffffff;
```

```
  border-color: #312e81;
}

.no-doc-tag {
  display: block;
  text-align: center;
  font-size: 12px;
  color: #94a3b8;
  font-style: italic;
  padding: 6px 0;
}

.auth-viewport {
  min-height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
  background: linear-gradient(135deg, #1e1b4b 0%, #312e81 50%, #4338ca 100%);
  padding: 20px;
}

.auth-card {
  background: #ffffff;
  border-radius: 16px;
  width: 100%;
  max-width: 400px;
  padding: 36px 30px;
  box-shadow: 0 20px 40px rgba(0, 0, 0, 0.25);
  border: 1px solid rgba(255, 255, 255, 0.2);
}

.auth-header {
  text-align: center;
  margin-bottom: 24px;
}

.auth-logo-badge {
  display: inline-block;
  background-color: #312e81;
  color: #06b6d4;
  font-size: 14px;
  font-weight: 900;
  padding: 8px 14px;
  border-radius: 8px;
  letter-spacing: 1px;
  margin-bottom: 12px;
}

.auth-header h2 {
  font-size: 21px;
  font-weight: 700;
  color: #1e1b4b;
  margin-bottom: 4px;
```

```
}

.auth-header p {
  font-size: 13px;
  color: #64748b;
}

.field-group {
  margin-bottom: 16px;
}

.field-group label {
  display: block;
  font-size: 13px;
  font-weight: 700;
  color: #334155;
  margin-bottom: 6px;
}

.field-group input {
  width: 100%;
  padding: 11px 13px;
  border: 1px solid #cbd5e1;
  border-radius: 8px;
  font-size: 13.5px;
  outline: none;
  transition: border-color 0.2s, box-shadow 0.2s;
}

.field-group input:focus {
  border-color: #4f46e5;
  box-shadow: 0 0 0 3px rgba(79, 70, 229, 0.15);
}

.auth-submit-btn {
  width: 100%;
  padding: 12px;
  background: linear-gradient(135deg, #312e81 0%, #4338ca 100%);
  color: #ffffff;
  border: none;
  border-radius: 8px;
  font-size: 14px;
  font-weight: 700;
  cursor: pointer;
  transition: opacity 0.2s;
  margin-top: 8px;
}

.auth-submit-btn:hover {
  opacity: 0.95;
}

.notice-banner {
```

```
  padding: 10px 14px;
  border-radius: 8px;
  font-size: 12.5px;
  margin-bottom: 16px;
  text-align: center;
  font-weight: 600;
}

.notice-danger {
  background-color: #fef2f2;
  color: #991b1b;
  border: 1px solid #fecaca;
}

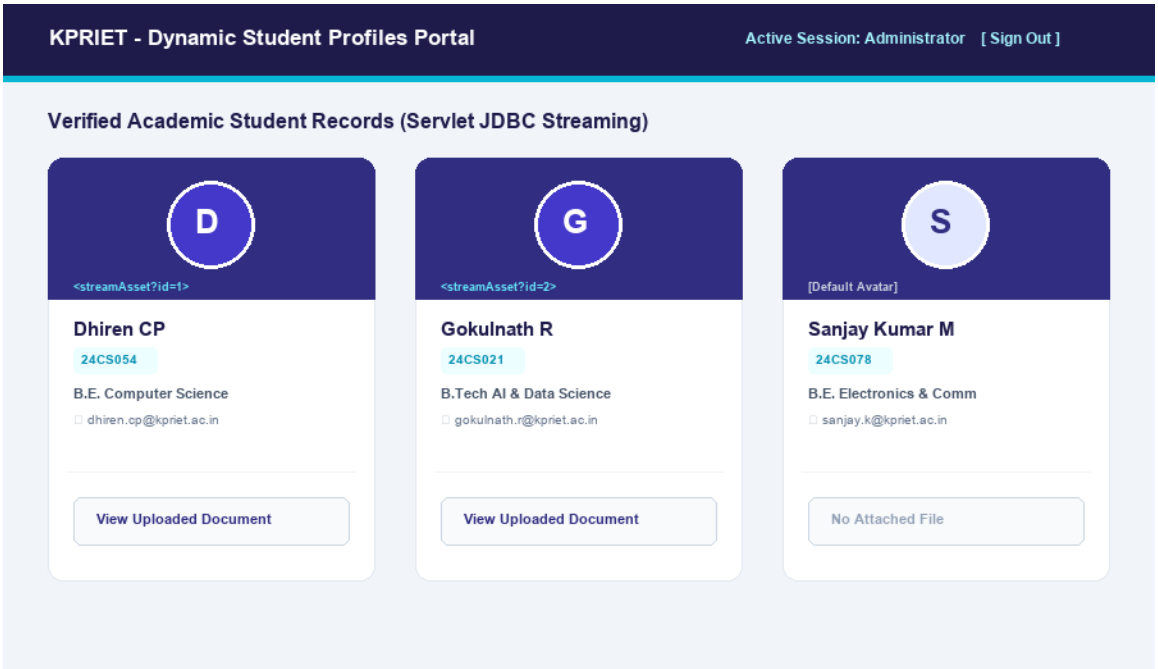
.notice-success {
  background-color: #ecfeff;
  color: #0e7490;
  border: 1px solid #a5f3fc;
}

.auth-card-footer {
  margin-top: 24px;
  text-align: center;
  font-size: 11.5px;
  color: #94a3b8;
  border-top: 1px solid #f1f5f9;
  padding-top: 14px;
}
```

OUTPUT :

VALID OPERATIONS :

Valid Operation – Verified Student Profile Dossiers with Servlet Streamed Assets :



RUBRICS:

Rubrics	**Actual Marks**	**Awarded Marks**
Servlet Architecture & web.xml Mapping (R1)	4	
JDBC Data Retrieval & Resource Management (R2)	4	
Servlet Binary Stream Handling (R3)	4	
Dynamic JSP Rendering & Conditional Logic (R4)	4	
Session Security & Authentication (R5)	4	
Total Marks	**20**	

RESULT:

A dynamic Java Web Portal was successfully developed and tested using Java Servlets, JDBC, and JSP. The system connects to MySQL via JDBC, streams binary photo assets directly to <img> tags using AssetStreamServlet, dynamically iterates over student profile records with JSP scriptlets, manages file attachment links conditionally, and enforces robust session security for administrative access.